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CPE 3101L

Pre-Midterm Exam for CO1

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1  /*****  
2  * AUTHOR:      CYRIL ANDRE B. DURANGO  
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL  
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM  
5  *  
6  * PROJECT FILE TITLE: ProblemCO1.v  
7  *****/  
8  
9  module ProblemCO1(G,A,B,C);  
10  
11     input    A, B, C;  
12  
13     output   G;  
14  
15     wire     w1, w2, w3, w4;  
16  
17     not      N1(w1, C);  
18  
19     or       OR1(w2, B, C);  
20     or       OR2(G, w3, w4);  
21  
22     and      AND1(w3, B, w1);  
23     and      AND2(w4, A, w2);  
24  
25 endmodule
```

Figure 1. Verilog Description File

Quartus Prime Lite Edition - Z:/Pre-Midterms/ProblemCO1/ProblemCO1 - ProblemCO1

File Edit View Project Assignments Processing Tools Window Help

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Project Navigator Hierarchy

Entity Instance

MAX 10: 10M02DCU324A6G

ProblemCO1

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Flow Summary

Flow Status: Successful - Fri Sep 27 08:22:08 2024

Quartus Prime Version: 20.1.1 Build 720 11/11/2020 SJ Lite Edition

Revision Name: ProblemCO1

Top-level Entity Name: ProblemCO1

Family: MAX 10

Device: 10M02DCU324A6G

Timing Models: Final

Total logic elements: 1

Total registers: 0

Total pins: 4

Total virtual pins: 0

Total memory bits: 0

Embedded Multiplier 9-bit elements: 0

Total PLLs: 0

UFM blocks: 0

ADC blocks: 0

Messages

Type ID Message

- 286030 Timing-Driven Synthesis is running
- 16010 Generating hard_block partition "hard_block:auto_generated_inst"
- 21057 Implemented 5 device resources after synthesis - the final resource count might be different
- Quartus Prime Analysis & Synthesis was successful. 0 errors, 1 warning
- Running Quartus Prime Netlist Viewers Preprocess
- Command: quartus_npp ProblemCO1 -c ProblemCO1 --netlist_Type=sgate
- 18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate value for
- Quartus Prime Netlist Viewers Preprocess was successful. 0 errors, 1 warning

System (1) Processing (16)

100% 00:00:02

Figure 2. Compilation Report - Flow Summary of ProblemCO1.v

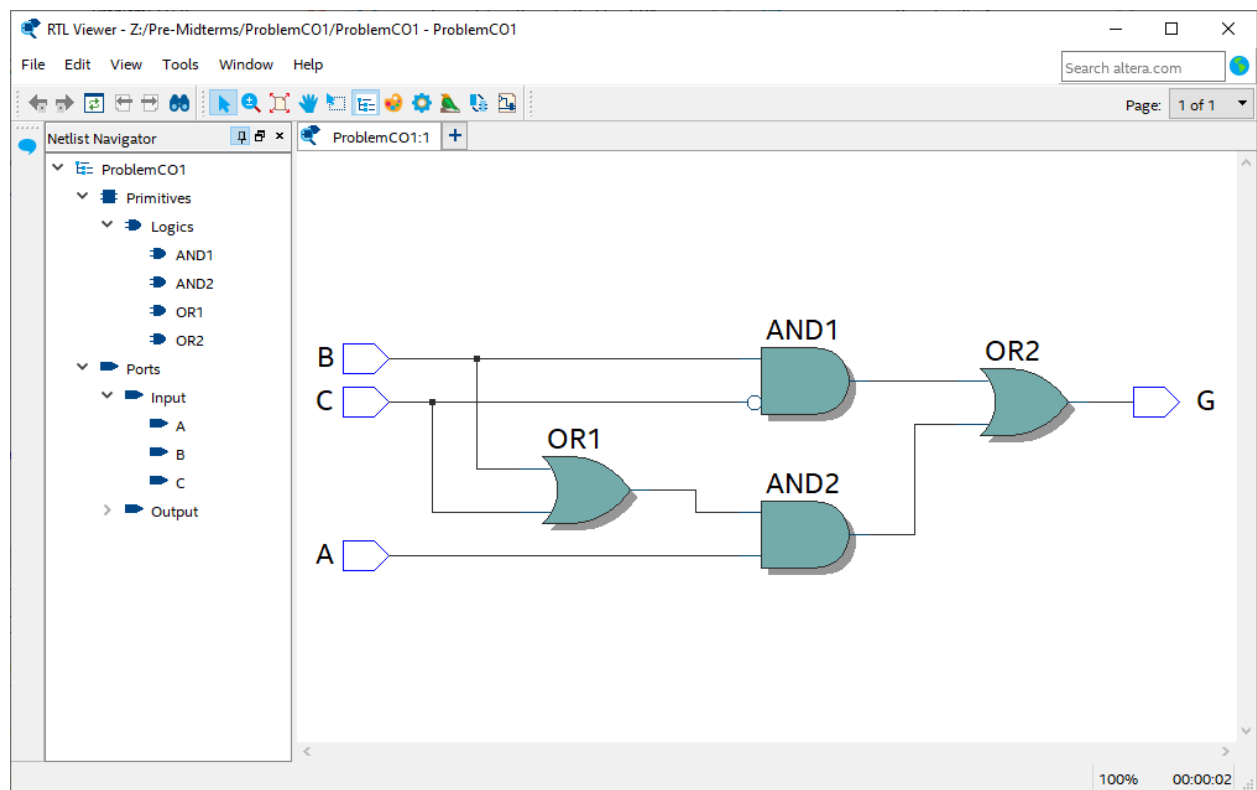


Figure 3. RTL Schematic Diagram of ProblemCO1.v